



Department of Computer Science and Engineering

Academic Year 2022– 2023(Odd Semester)

Degree, Semester & Branch: III Semester B.E. ECE-‘A’ Section

Course Code & Title: C Programming and Data Structures

Name of the Faculty member (s): Dr.V. Anusuya, ASCP/CSE

Innovative Practice Description

- **Unit / Topic:** Unit 3 / Applications
- **Course Outcome:** The student will be able to Implement linear data structure operations using functions.
- **Topic Learning Outcome:** TLO9.
- **Activity Chosen:** Think Pair Share
- **Justification:**
 - After explaining the concept of evaluation of expression such as infix to postfix conversion and postfix evaluation in stack application, to understand the learning level of the student about evaluating the expression this activity was given.
- **Time Allotted for the Activity:** 15 Minutes
- **Details of the Implementation:**
 - Displayed the different questions in the topics infix to postfix conversion and postfix evaluation for each pair of students in the class as shown in Figure 1.
 - Identified 32 pair of students and asked them to think for five minute about evaluation of expression.
 - After that, the students were asked to discuss the answer for the questions in the Dias as shown in the figure 2,3.
- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO5	PO9	PO10	PO12	PSO1
CO3	3	3	2	2	1	1	2	1	2

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO1	PO2	PO3	PO4	PO5	PO9	PO10	PO12	PSO1
Justification for correlation	3	3	2	2	1	1	2	1	2
	The basic knowledge of engineering fundamentals is necessary to	The student should identify analyze	The student will be able to design	The student will be able to find the problem for research	The student will be able to use modern tool they	The student doing the problem by the	The students improve communication by sharing knowledge in the	The application of stack is necessary for lifelong	The student will be able to apply stack application in the field of



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	understand the concept of expression evaluation	problem to evaluate expression	and find the solution for the problem	problem they need knowledge about expression evaluation	need knowledge of evaluation of expression	group of two members	form of presentation	learning	information and Communication Technology
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• Images / Screenshot of the practice:

Applications of Stack

Q.No.1: Convert the following infix expression to its equivalent postfix expression. Show stack contents for the conversion
 $(A+B)(C-D)/E$

Q.No.2: Change the following infix expression into postfix expression
 $(A+B/C)+D/E-F$

Q.No.3: Obtain the postfix notation for the following infix notation of expression showing the contents of the stack and postfix expression formed after each step of conversion.
 $A*B/(C-D)^2$

Q.No.4: Conversion Of Infix Into Postfix
 $2*(4-5/6)$

Q.No.5: Change the following infix expression postfix expression
 $(A + B/C)*D/E-F$

Q.No.6: Evaluate the Postfix expression
 $AB-CD+E^*$ (where $A=5, B=3, C=5, D=4$, and $E=2$)

Q.No.7: Convert infix to postfix expression
 $A/B^*C-D^*E-A^*C$

Q.No.8: Write the postfix for the following
 A^*B^*C

Q.No.9: Write the postfix for the following
 $-A/B-C/D$

Q.No.10: Write the postfix for the following
 A^*B-C

Q.No.11: Evaluate the following postfix expression
 $5.4+9.3/-$

Q.No.12: Write the postfix for the following
 $((A/B-C+D)/E-A)^*C$

Q.No.13: Write the postfix for the following
 $A/B-C/D^*E-A^*C$

Q.No.14: Evaluate the following postfix expression
 22^*14^*

Q.No.15: Write the postfix for the following
 A^*B+C

Q.No.16: Write the postfix for the following
 $A/B-C/D^*E-A^*C$

Q.No.17: Evaluate the following postfix expression
 $12+4/3^*2-5$

Q.No.18: Write the prefix for the following
 $A^*(B+C)/D-G$

Q.No.19: Evaluate the following postfix expression
 $1.6+2.3^*$

Q.No.20: Write the prefix for the following
 $(A+B)^*(D+E)/(F+A^*D)+C$

Q.No.21: Evaluate the following postfix expression
 $100/200+2/5^*7-$

Q.No.22: Evaluate the following postfix expression
 $5.3+8.2^*$

Q.No.23: Write the prefix for the following
 $A/B^*C+D^*E-A^*C$

Q.No.24: Evaluate the following postfix expression
 $abc^*+da^*b^*c^*$, where $a=1, b=2, c=3, d=4, e=5, f=6, g=2$.

Q.No.25: Evaluate the following postfix expression
 A^*B-C/D

Q.No.26: Evaluate the following postfix expression
 $12+3*54-(3^*56)+7.8.3-(4/5)^*(1^*5)+6$

Q.No.27: Write the postfix for the following
 $a+b^*c+(d^*e+fg)^*$

Q.No.28: Write the prefix for the following
 $(abc)^*(cda)$

Q.No.29: Evaluate the following postfix expression
 $54.6+7.4-8.9/5.5.15+9$

Q.No.30: Evaluate the following expression after converting postfix.
 $(abc)^*(cda)$ where $a=1, b=2, c=3, d=4$

Q.No.31: Write the postfix for the following
 $((A+B)^*(D^*E)-F)$

Q.No.32: Evaluate the following postfix expression
 $1.2.3^*+4-$

Figure 1 Questions given to students

Screen shots

Q) Write the postfix for the following
 $((A/B-C+D)/E-A)^*C$

Input	Stack	Output	Remarks
(	[(]		
(	[((]		
A	[((A]	A	
/	[((/]	/	
(	[((/ (]		
B	[((/ (B]	AB	
-	[((/ (-]	AB-	
C	[((/ (- C]	ABC-	
+	[((/ (- +]	ABC-	
D	[((/ (- + D]	ABC-D	
)	[((/ (- +)]	ABC-D+	

[(]	ABC-D+/
[(/]	ABC-D+/
[(/ (]	ABC-D+/
[(/ (B]	ABC-D+/E
[(/ (-]	ABC-D+/E
[(/ (- C]	ABC-D+/EA
[(/ (- +]	ABC-D+/EA-
[(/ (- + D]	ABC-D+/EA-C
[(/ (- +)]	ABC-D+/EA-C^*
[(/ (- +) C]	ABC-D+/EA-C^*
[(/ (- +) C]	ABC-D+/EA-C^*



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$a + (b * c + (d * e + f) * g)$			At: Afreen, Gaurav B. Ranjita, Varunika.		
IP	Stack		OP		
a			a		
+	[+]		a		
b			ab		
*	[*]		ab*		
c			ab*c		
+	[+]		ab*c+		
d			ab*c*d		
*	[*]		ab*c*d		
e			ab*c*d*		
+	[+]		ab*c*d*+		
f			ab*c*d*+		
)			ab*c*d*+		
*			ab*c*d*+		
g			ab*c*d*+		

Figure 2. Sample Answers of students



Figure 3. Presentation of students

- **Reflective Critique:**

- ***Feedback of practice from students and other stakeholders:***
 - The students are actively participated in the activity.
 - The student said this activity helped to think and recall information and improved our problem solving skills.
- ***Benefit of the practice:*** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)
 - The majority of students correctly answered the question that was given to them.
 - During class, there was effective interaction between the students and the teacher.
 - When students communicated to their peers, their nervousness was reduced.
 - It helps the students to think individually about a topic or answer to a given question.
 - With the help of this activity, each student can review the idea of evaluation of expression and infix to postfix conversion and gain a thorough knowledge of the topic.

- ❖ ***Challenges faced in implementation:***

Most of the students are hesitant to discuss the ideas they have learned in the classroom.

References:

- <https://www.ritrjpm.ac.in/images/computer-science/>

Signature of Faculty Member

HOD